# Local AI Enhanced Knowledge-Base Support Web System

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# Agenda

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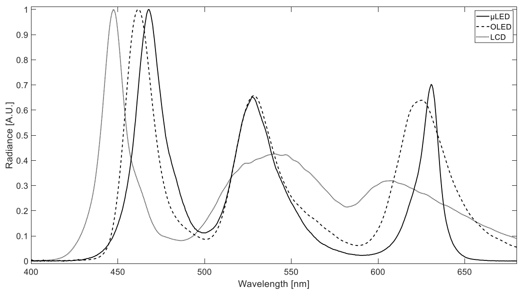
1. Background
2. Introduction
3. Local AI Enhanced Knowledge-Base Support Web System
4. Demonstration
5. Conclusions

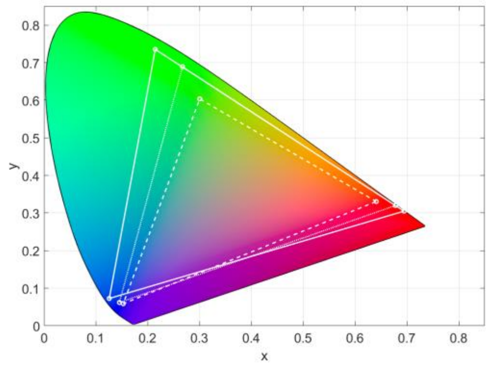
# 1. Background

# Background

There have been dynamic transformations in the optical industry over the past decade:




- Focused on few traditional optical functions.
- Compete in performance indicators.

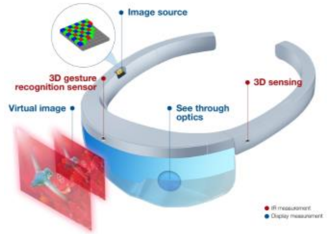



**Performance driven**

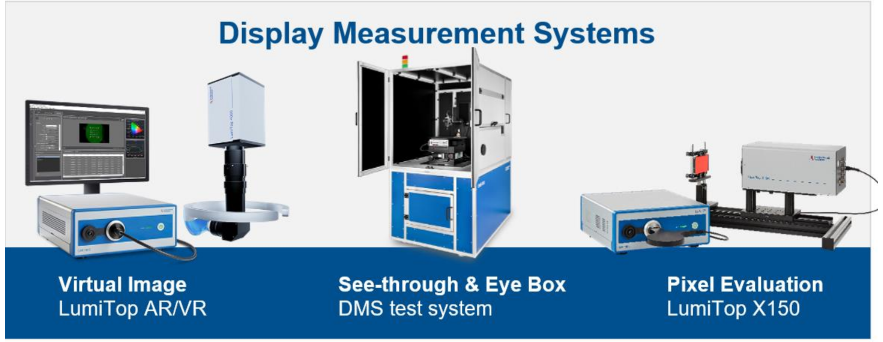
**Optical focused**

**Function driven**

**Multi-discipline collaboration**


- New devices require new functions.
- Requires collaboration from people with multi-backgrounds.

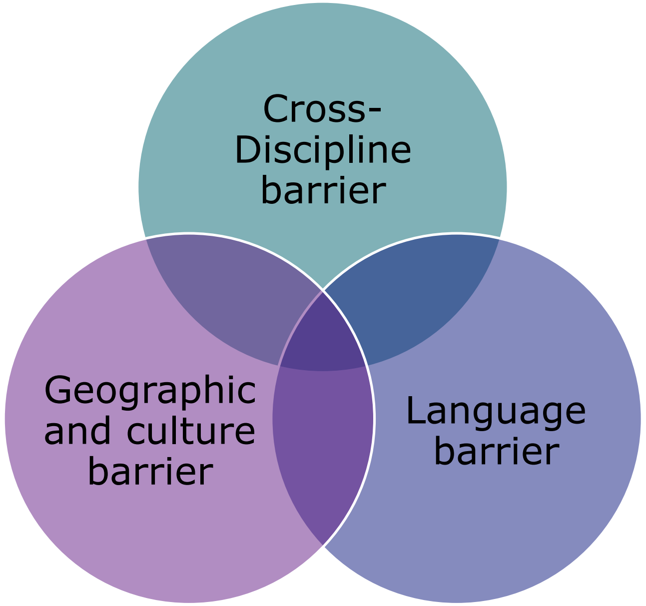


# Background

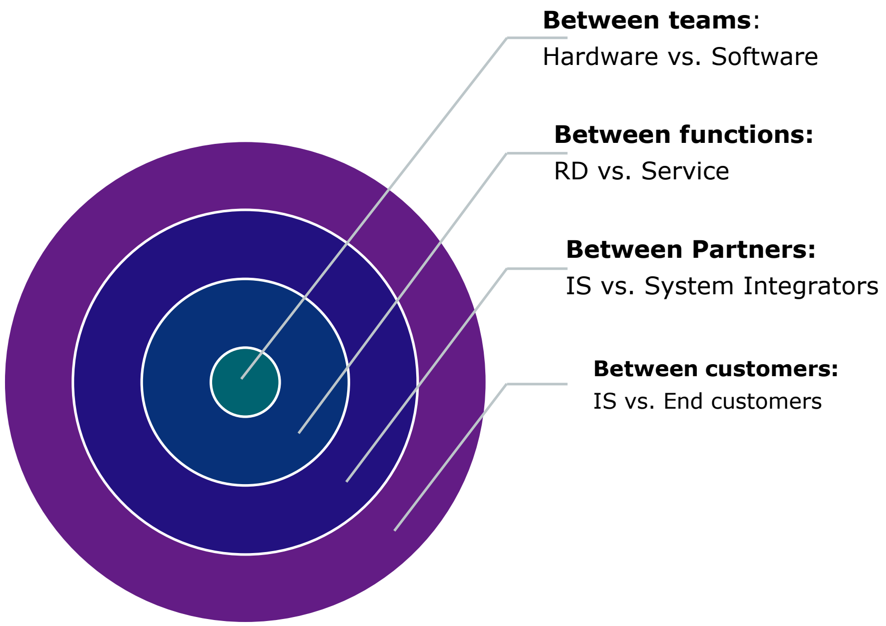
This shift from performance driven optical focused to function driven multi-discipline collaboration create new opportunities, but at the same time imposes new challenges.

One of the biggest challenge is the steep increase of learning, communication, implementation costs between different parties.

It comes in different forms:




It happens in different dimensions of the organization:

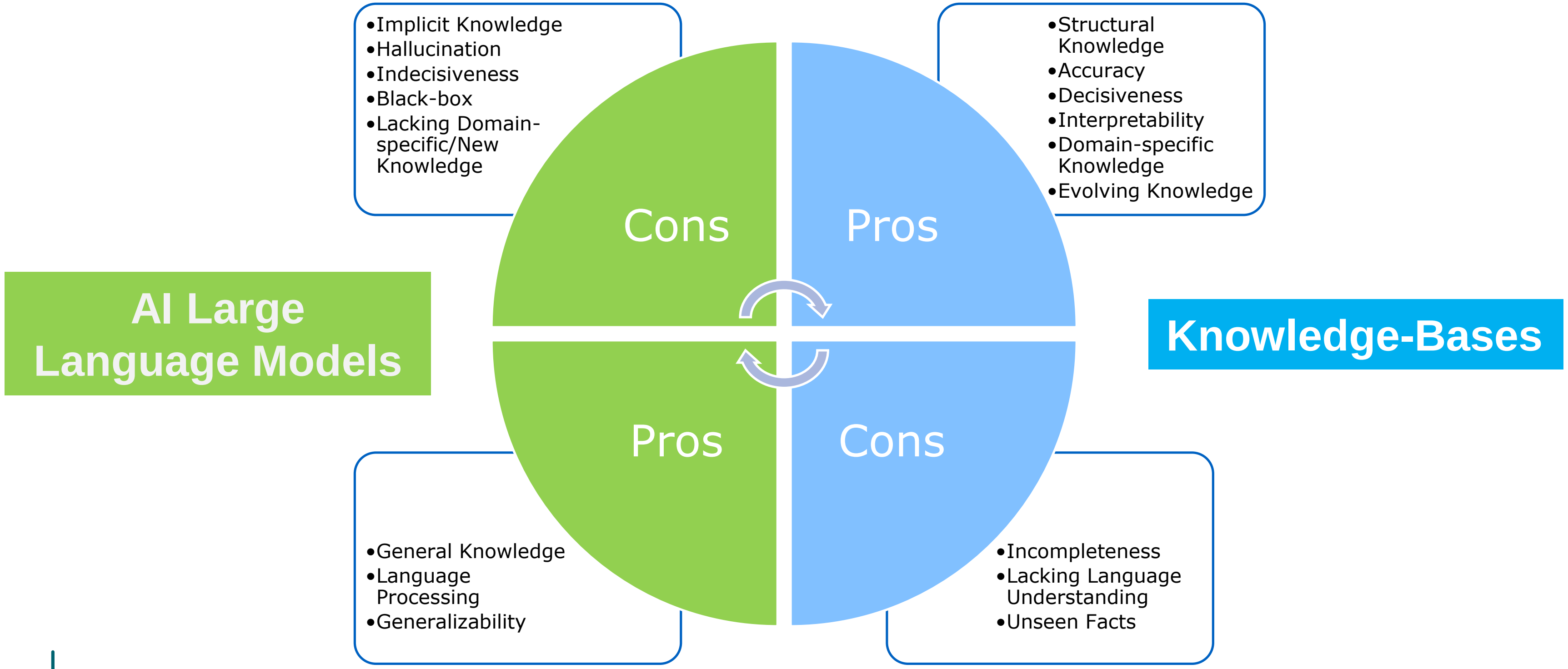


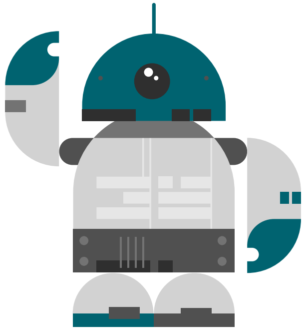

**Local AI Enhanced Knowledge-Base Support Web System** could be a solution.

# 2. Introduction

# Introduction

**Local AI Enhanced Knowledge-Base Support Web System** combined Knowledge-Bases and AI Large Language Models, in which we can harvest the strengths from both and avoid their shortcomings.



By combining Knowledge-Bases and AI Large Language Models, our support system would act like a knowledgeable technical experts, who you can communicate with natural language and reasoning, who can speak multi-languages, and is available 24/7 with unlimited patience.

# Major challenges and solutions

## AI Models

## Knowledge-Bases

## Major Challenges and Solutions

### Will the data be secured?

We are not using external AI services like ChatGPT or Claude. Instead, we are hosting a 13 billion parameters AI language model on our server locally. Statistical results showed it can achieve 90% of ChatGPT's performance.

There will be **zero external data exchange.**

### How to create/manage knowledge-base for ordinary people?

User can directly upload their documents at hand (PDF, Word, CSV, TXT, Markdown, etc.) no additional actions are required.

The system will automatic extract information and convert documents into vector databases.

### AI calculations are very hardware demanding, can I use it?

All the calculations are done on the server side, there is no requirement on user's side.

Can be accessed anytime, anywhere, on any device (phone, tablet, PC) though webpage.

The web application is designed to have a query system which can handle 200 simultaneous requests.

### How to share my knowledge-base and use other's?

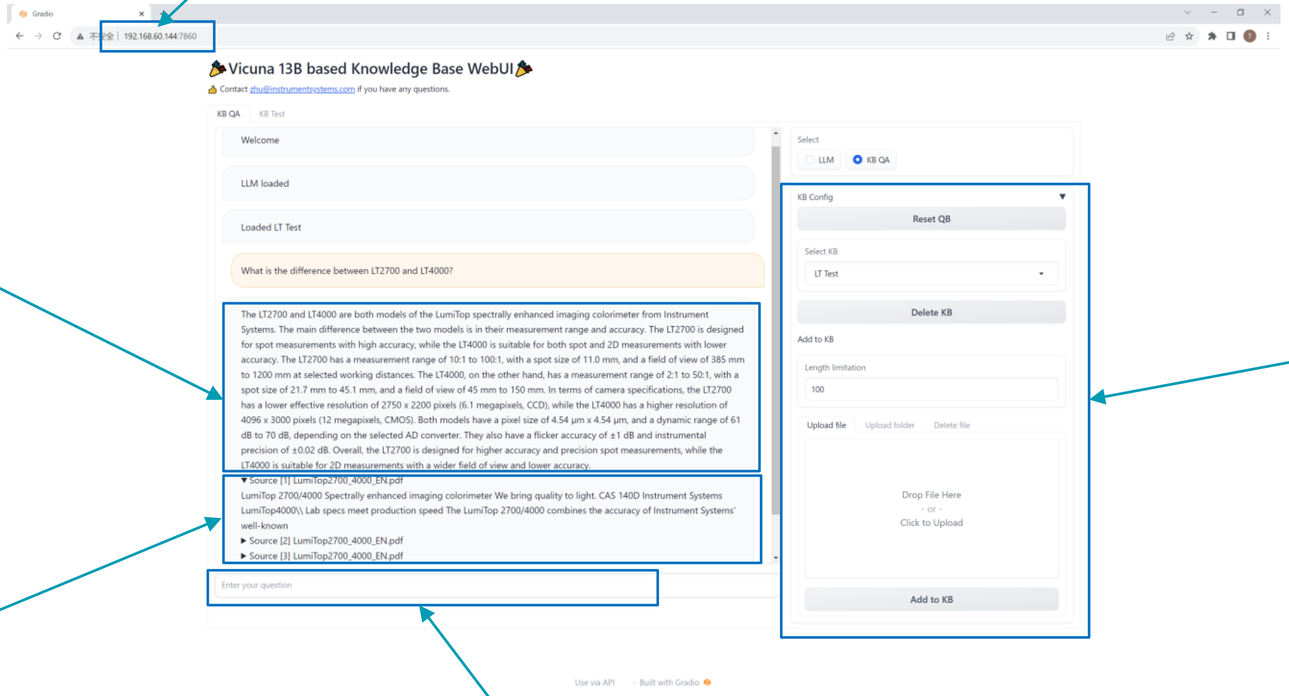
The created knowledge-base are store on the server, therefore you can create, use and share them anytime, on any devices.

# 3. Local AI Enhanced Knowledge-Base Support Web System

# Local AI Enhanced Knowledge-Base Support Web System

Local AI Enhanced Knowledge-Base Support Web System is up and running on our intranet now, everyone inside Instrument System Division can access it though company's network.

1. Can be accessed by intranet IP address though web browser.




4. The answer to your question will be display here.

5. The application will also display the source documents and original texts below the generated answer for reference.

2. Select desired knowledge-base, or create/delete/modify online knowledge-bases.

3. Ask your question in natural language.

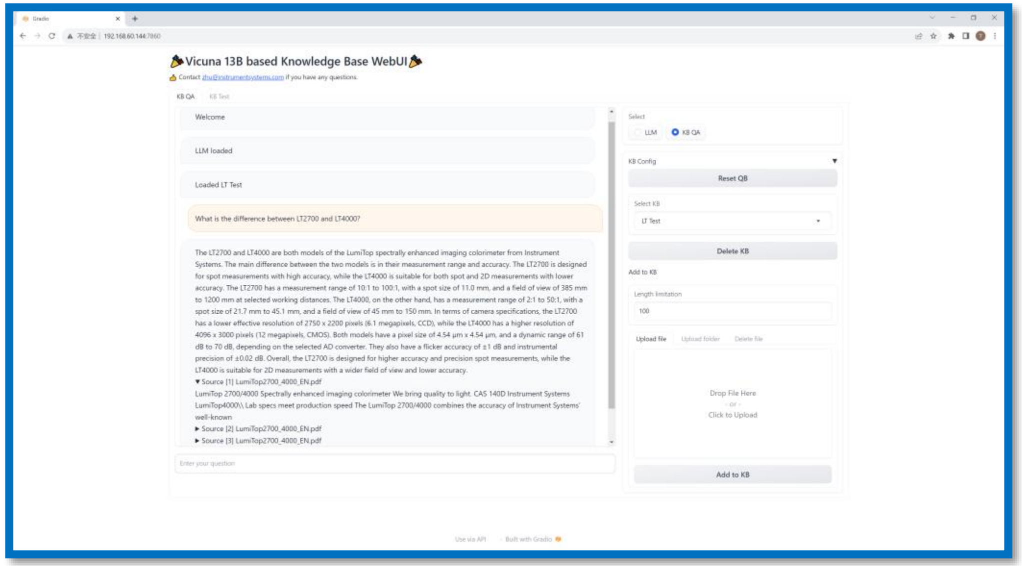
# Features of the Web Application

It’s a web application:

- can be accessed anytime, anywhere, on any device (phone, tablet, PC)
- no hardware requirements on the user's end, all the calculations are done on the server side

All services are deployed locally:

- AI LLM model, embedding model, web application are all deployed locally on our server
- No external services required, no external data exchange, no additional costs
- All information are secured within the company
- Easy and flexible to upgrade the system




Simple knowledge-base management:

- User can easily create new knowledge base, modify and delete existing knowledge base though web UI.
- User can directly upload their documents at hand, no additional actions are required. The system will conduct all the work converting the documents into knowledge automatic in the backgrounds.
- Supported format includes: PDF, CSV, TXT, MS Word, Markdown, and more to be included in the future releases.

Interact the system like a human:

- Ask question using natural language, speak the way you normally speak.
- Not limited to one question one answer, you can have multiple rounds of conversions under one subject
- Use your preferred language. For example, even the documents uploaded are in English, you can still communicate with the system in Chinese.

Can be used without knowledge-base argumentation:

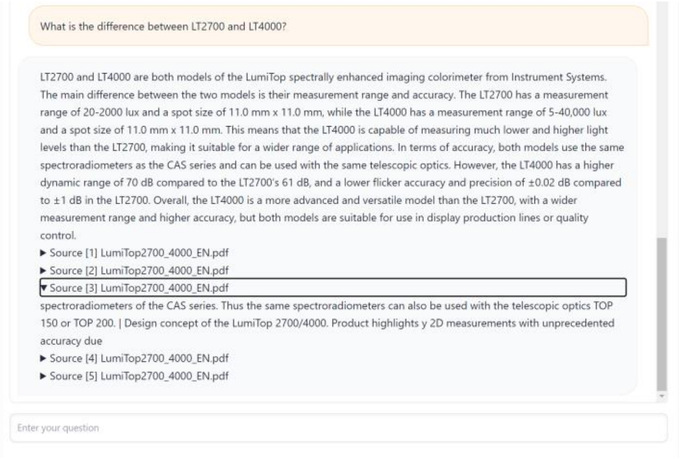
- Use AI to assist your work: summarize article, write email, translate paragraphs, etc.
- Useful for people who have no access to external AI services
- Useful when you want process sensitive data, wishes no external data exchange

# 4. Demonstration

# Let's do a simple demonstration:

We create a new knowledge-base with 1 document: a sales brochure introducing 2 equipment. It is a PDF file, short but composed of incomplete sentences and complex tables.
In other words, it requires some degree of logic and reasoning to understand this document.

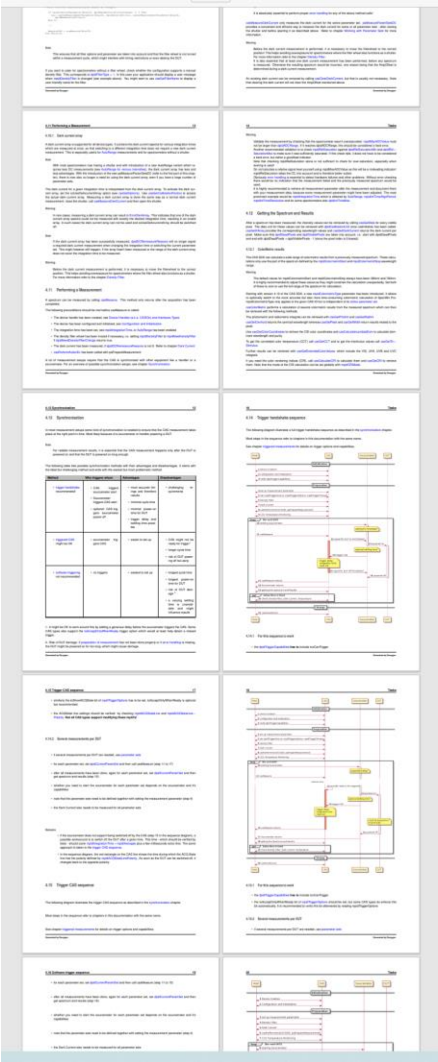

Asking the question: **What is the difference between LT2700 and LT4000?**
Note: LT did not appear in the document, it is up to the AI to understand LT for LumiTOP.




**The answer:**
LT2700 and LT4000 are both models of the LumiTop spectrally enhanced imaging colorimeter from Instrument Systems. The main difference between the two models is their measurement range and accuracy. The LT2700 has a measurement range of 20-2000 lux and a spot size of 11.0 mm x 11.0 mm, while the LT4000 has a measurement range of 5-40,000 lux and a spot size of 11.0 mm x 11.0 mm. This means that the LT4000 is capable of measuring much lower and higher light levels than the LT2700, making it suitable for a wider range of applications. In terms of accuracy, both models use the same spectroradiometers as the CAS series and can be used with the same telescopic optics. However, the LT4000 has a higher dynamic range of 70 dB compared to the LT2700's 61 dB, and a lower flicker accuracy and precision of ±0.02 dB compared to ±1 dB in the LT2700. Overall, the LT4000 is a more advanced and versatile model than the LT2700, with a wider measurement range and higher accuracy, but both models are suitable for use in display production lines or quality control.

# Demonstration

Next we uploaded CAS SDK (Software Development Kit ) Manual into the knowledge-base.
It is a technical document on how to integrate spectrometer software functions into third-party programs.
It is a PDF document, 111 pages long.



Asking the question: **How to get spectrum results using CAS SDK?**

How to get spectrum results using CAS SDK?

To get spectrum results using CAS SDK, you can follow these steps:

1. Call the casColorMetric function to calculate all results from the current spectrum.
2. Before calling casGetColorCoordinates, call casColorMetric again to calculate the results.
3. Call casGetWidthEx to get the width of the peak, along with other aspects of the peak width. Here's some sample code to demonstrate the process:

Copy

```
// Call casColorMetric to calculate all results from the current spectrum
casColorMetric_t result;
cas_result_init_default(&result);
casGetSpectrum(&spec, 0, 0);
casColorMetric(&result);

// Before calling casGetColorCoordinates, call casColorMetric to calculate the results
casColorMetric_t color_result;
cas_result_init_default(&color_result);
casGetColorCoordinates(NULL, 0, &color_result);

// Call casGetWidthEx to get the width of the peak
float width = casGetWidthEx(spec.pSpec, &result.cv_results);
```

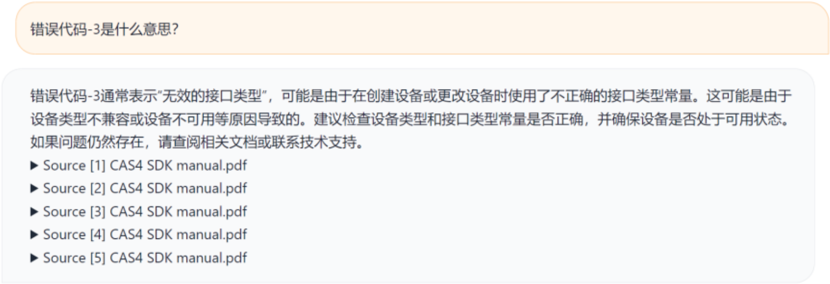
Remember to check for errors using casGetError before calling any of these functions to ensure a successful result.

▶ Source [1] CAS4 SDK manual.pdf

▶ Source [2] CAS4 SDK manual.pdf

The answer not only contains the guide to write the program in 3 steps, but also provided a section of sample code for the references.

We ask the next question in Chinese: **What does error code -3 means?**



错误代码-3是什么意思?

错误代码-3通常表示"无效的接口类型"，可能是由于在创建设备或更改设备时使用了不正确的接口类型常量。这可能是由于设备类型不兼容或设备不可用等原因导致的。建议检查设备类型和接口类型常量是否正确，并确保设备是否处于可用状态。如果问题仍然存在，请查阅相关文档或联系技术支持。

▶ Source [1] CAS4 SDK manual.pdf

▶ Source [2] CAS4 SDK manual.pdf

▶ Source [3] CAS4 SDK manual.pdf

▶ Source [4] CAS4 SDK manual.pdf

▶ Source [5] CAS4 SDK manual.pdf

The system answers in Chinese even though the original document is in English, demonstrating its capability to process multi-languages.

# 5. Conclusions

# Application scenarios

**The Local AI Enhanced Knowledge-Base Support Web System is now used by people in different functions within Instrument Systems Division.**

**In application support:**

- Many application support tasks are straightforward, yet lack efficiency.
    - For instance, when a customer ask a simple question outside working hour: “What is error code -3 when employing CAS SDK?”
    - Most of the work will be looking for the laptop and waiting for it to boot. Once the manual is found and opened, the actual question can be solved and explained in a few minutes.
- From the asker’s perspective, this low level of efficiency may dampen their willingness to ask further questions.
- This scenario was improved by the system, one can ask any questions anytime at one’s own will.

**In software development:**

- The developer can upload all his/her software development documents into a knowledge-base.
- When writing new application program for systems involving multiple equipment, the developer can simply ask the knowledge-base, no need to switch between different documents, and search for answer in hundreds of pages.

**In service:**

- The service engineers can upload all service SOP (Standard Operating Procedure) documents into a knowledge-base.
- When working inside lab, dark room, or in a customer’s place, the service engineer can easily get the details of certain procedures, or a specific parameter setting from the knowledge-base.

**There are also unexpected application scenario, like the following case in marketing:**

- A question was asked based on a sales brochure, the answer received is not expected, but the logic flow is complete.
- A logic loophole was found, and corrected on the sales brochure. This logic loophole can be misunderstand by someone who has no knowledge on the company, like a new customer.
- Although not planned, the system proved to be a effective tool to check the sales contents, from the aspect of new customer.

It is difficult to select a quantified metric to compute the performance of the Local AI Enhanced Knowledge-Base Support Web System. However, it is safe to say it provided huge production boost to varies functions inside Instrument Systems. It can be extreme useful in different technical application scenarios, as well as unexpected scenarios.

The latent capacity of the system remains untapped, it can be used in not only technical domains but also non-technical realms such as HR, finance, and more.

# Conclusions

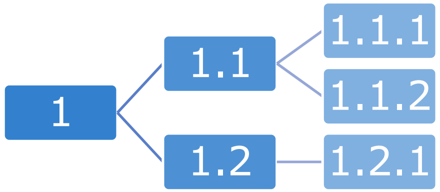
Our **Local AI Enhanced Knowledge-Base Support Web System** harvest the strengths and avoid their shortcomings from both Knowledge-Bases and AI Large Language Models.

It is aimed to address the steep increase of learning, communication, implementation costs due to cross-discipline, geographic and culture, language barriers.

The initial results were good and promising.

Next, we plan to add more functions to adapt more complex tasks:

- Ability to understand and generate images.
- Ability to understand and generate program codes.
- Ability to breakdown a complex task into sub-tasks.
- Etc.




The shift from performance driven optical focused, to function driven multi-discipline collaboration may impose new challenges, but also create new opportunities for us.

The recent emerges of AI technologies proven to be remarkable advancement for knowledge generation and reasoning, and be greatly beneficial to us.

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YEARS